

Supplementary Table 1. OPA geometric mean titers before and after the PCV13 booster dose (per protocol set)

Serotype	3+1 schedule (Czech Republic)				2+1 schedule (Slovakia)			
	PHiD-CV group N = 53		PCV13 group N = 45		PHiD-CV group N = 39		PCV13 group N = 49	
	N	GMT (95% CI)	N	GMT (95% CI)	N	GMT (95% CI)	N	GMT (95% CI)
Serotype 1								
Month 0	51	5.6 (4.5; 6.9)	44	5.7 (4.5; 7.1)	37	7.7 (5.4; 10.8)	49	6.8 (5.3; 8.8)
Month 1	51	287.7 (213.7; 387.3)	45	177.6 (135.1; 233.4)	38	291.8 (197.2; 431.8)	49	226.8 (161.7; 318.0)
Month 12	46	11.0 (7.6; 15.8)	44	5.7 (4.3; 7.7)	37	11.4 (7.6; 17.3)	48	11.6 (8.5; 15.7)
Serotype 3								
Month 0	50	4.9 (3.8; 6.2)	44	7.5 (5.2; 10.8)	38	4.4 (3.9; 4.9)	46	11.7 (8.1; 16.8)
Month 1	51	205.5 (164.5; 256.6)	45	128.0 (92.3; 177.6)	38	135.1 (93.5; 195.0)	49	135.4 (105.3; 174.2)
Month 12	40	12.5 (8.6; 18.3)	43	7.9 (5.6; 11.1)	32	14.3 (9.1; 22.5)	40	16.3 (10.6; 25.1)
Serotype 4								
Month 0	49	72.3 (46.1; 113.3)	43	60.0 (33.0; 109.1)	35	43.8 (23.1; 82.9)	47	43.2 (27.2; 68.5)
Month 1	51	1691.3 (1366.1; 2093.9)	45	3455.1 (2672.6; 4466.7)	38	1372.1 (1067.6; 1763.4)	49	1995.2 (1598.0; 2491.2)
Month 12	46	44.5 (25.1; 79.0)	43	112.2 (57.6; 218.6)	36	31.1 (15.1; 64.4)	48	86.8 (52.4; 143.7)
Serotype 5								
Month 0	48	21.2 (13.6; 33.1)	43	22.2 (16.3; 30.4)	36	32.2 (19.6; 53.0)	45	25.4 (18.0; 35.7)
Month 1	51	671.2 (528.4; 852.5)	45	477.1 (376.7; 604.2)	38	733.7 (528.5; 1018.7)	49	470.4 (377.6; 586.1)
Month 12	46	72.8 (50.0; 106.0)	34	38.4 (24.1; 61.0)	35	69.3 (45.4; 105.8)	48	40.1 (28.0; 57.4)
Serotype 6A								
Month 0	49	32.1 (15.3; 67.5)	44	525.5 (349.2; 790.9)	36	34.6 (15.3; 77.9)	47	119.9 (70.5; 203.8)
Month 1	51	4017.0 (2708.5; 5957.8)	45	8807.5 (7174.8; 10811.6)	38	6389.4 (4690.2; 8704.2)	49	5991.7 (3965.5; 9053.2)
Month 12	46	671.0 (414.2; 1087.1)	44	1097.4 (837.5; 1438.1)	38	710.6 (402.0; 1256.2)	49	966.7 (677.0; 1380.4)
Serotype 6B								
Month 0	50	408.6 (307.9; 542.3)	45	146.1 (81.4; 262.2)	36	358.5 (200.4; 641.3)	46	43.8 (21.5; 89.6)
Month 1	51	6289.5 (4807.6; 8228.3)	45	5577.0 (4388.4; 7087.6)	38	6933.8 (5165.6; 9307.2)	49	4332.1 (3203.6; 5858.1)
Month 12	48	657.3 (466.0; 927.0)	43	532.4 (348.2; 814.0)	38	389.2 (224.5; 674.6)	49	236.5 (144.3; 387.7)
Serotype 6C								
Month 0	49	24.3 (11.6; 50.8)	42	189.3 (92.8; 386.3)	36	23.4 (10.3; 53.1)	46	65.0 (31.0; 136.2)
Month 1	51	3207.6 (2164.0; 4754.7)	45	10499.0 (8059.5; 13676.9)	38	1171.8 (602.7; 2278.1)	49	2680.1 (1535.4; 4678.3)
Month 12	41	635.1 (310.3; 1300.1)	42	1031.1 (707.0; 1503.7)	36	189.4 (73.4; 488.9)	45	457.5 (239.0; 875.7)
Serotype 7F								
Month 0	51	585.4 (455.8; 751.9)	45	797.6 (624.4; 1018.8)	37	721.8 (533.7; 976.2)	49	614.3 (486.8; 775.3)
Month 1	51	3130.4 (2605.1; 3761.6)	45	6281.4 (4528.6; 8712.5)	38	2792.3 (2229.7; 3496.8)	49	4801.4 (3698.7; 6232.8)
Month 12	47	901.6 (665.3; 1222.0)	45	1470.7 (1125.1; 1922.5)	38	857.1 (554.2; 1325.7)	49	1781.5 (1405.7; 2257.8)
Serotype 9V								
Month 0	49	212.9 (136.4; 332.2)	41	157.6 (84.4; 294.5)	38	43.6 (22.4; 84.9)	46	44.3 (27.4; 71.8)
Month 1	51	2078.7 (1664.4; 2596.1)	45	4099.2 (3032.7; 5540.8)	38	2305.9 (1797.8; 2957.5)	49	2815.0 (2161.9; 3665.4)
Month 12	44	537.0 (368.8; 781.9)	44	732.9 (525.8; 1021.6)	38	526.4 (336.6; 823.1)	47	679.6 (419.2; 1101.9)
Serotype 14								
Month 0	51	531.8 (373.0; 758.1)	45	612.2 (397.9; 941.8)	36	405.8 (266.1; 618.6)	49	340.7 (226.5; 512.4)
Month 1	51	1780.5 (1444.1; 2195.2)	45	3376.4 (2345.3; 4860.7)	38	1492.7 (1158.4; 1923.4)	49	2598.8 (1932.1; 3495.4)

Serotype	3+1 schedule (Czech Republic)				2+1 schedule (Slovakia)			
	PHiD-CV group N = 53		PCV13 group N = 45		PHiD-CV group N = 39		PCV13 group N = 49	
	N	GMT (95% CI)	N	GMT (95% CI)	N	GMT (95% CI)	N	GMT (95% CI)
Month 12	48	576.3 (436.0; 761.6)	44	706.9 (453.2; 1102.6)	31	259.3 (138.0; 487.2)	47	361.4 (256.1; 510.1)
Serotype 18C								
Month 0	51	185.4 (110.0; 312.5)	44	104.6 (59.9; 182.5)	37	102.3 (60.2; 173.8)	48	33.6 (21.3; 52.8)
Month 1	51	2072.9 (1680.9; 2556.4)	45	2552.4 (2002.4; 3253.5)	38	2122.1 (1577.2; 2855.2)	49	2270.4 (1741.1; 2960.5)
Month 12	47	836.6 (595.6; 1175.3)	43	772.7 (536.6; 1112.9)	37	490.0 (276.4; 868.8)	49	714.6 (443.1; 1152.4)
Serotype 19A								
Month 0	50	13.4 (8.6; 20.7)	45	65.1 (39.6; 107.1)	38	13.8 (7.6; 25.0)	46	41.5 (28.0; 61.6)
Month 1	51	2778.6 (2102.1; 3672.9)	45	3004.9 (2391.4; 3775.7)	38	1861.4 (1361.6; 2544.5)	49	2337.1 (1865.1; 2928.7)
Month 12	47	266.9 (150.0; 475.2)	44	355.8 (219.9; 575.6)	36	204.8 (117.1; 358.2)	48	292.0 (183.0; 465.9)
Serotype 19F								
Month 0	51	192.7 (122.4; 303.5)	43	47.5 (28.8; 78.3)	38	157.5 (100.6; 246.7)	48	41.1 (27.3; 61.9)
Month 1	51	3868.0 (3045.0; 4913.3)	45	2769.6 (2152.4; 3563.7)	38	2720.6 (2054.4; 3603.0)	49	2137.0 (1655.0; 2759.4)
Month 12	48	586.7 (427.2; 805.8)	39	267.5 (143.2; 499.5)	38	210.2 (118.7; 372.4)	49	180.5 (115.7; 281.8)
Serotype 23F								
Month 0	49	68.1 (36.3; 127.8)	41	159.0 (79.5; 317.7)	38	83.7 (41.1; 170.6)	46	44.5 (23.0; 85.9)
Month 1	51	3470.9 (2774.5; 4342.0)	45	6561.6 (4277.3; 10065.7)	38	3391.4 (2610.1; 4406.6)	49	5238.2 (3985.7; 6884.4)
Month 12	47	709.3 (464.7; 1082.9)	44	1338.3 (888.9; 2015.0)	36	283.3 (130.1; 616.6)	48	788.1 (454.1; 1367.6)

GMT, geometric mean titer; OPA, opsonophagocytic assay; PCV13, 13-valent

pneumococcal conjugate vaccine; PHiD-CV, 10-valent pneumococcal non-typeable

Haemophilus influenzae protein D conjugate vaccine

Supplementary Table 2. Percentage of subjects with OPA geometric mean titers ≥ 8 1 and 12 months after the PCV13 booster dose (per protocol set)

Serotype	Timing	3+1 schedule (Czech Republic)		2+1 schedule (Slovakia)	
		PHiD-CV group N = 51	PCV13 group N = 45	PHiD-CV group N = 38	PCV13 group N = 49
Serotype 1	Month 1	51 (100.0%)	45 (100.0%)	38 (100.0%)	49 (100.0%)
	Month 12	22 (43.1%)	6 (13.3%)	19 (50.0%)	30 (61.2%)
Serotype 3	Month 1	51 (100.0%)	45 (100.0%)	37 (97.4%)	49 (100.0%)
	Month 12	24 (47.1%)	15 (33.3%)	20 (52.6%)	30 (61.2%)
Serotype 4	Month 1	51 (100.0%)	45 (100.0%)	38 (100.0%)	49 (100.0%)
	Month 12	31 (60.8%)	32 (71.1%)	18 (47.4%)	40 (81.6%)
Serotype 5	Month 1	51 (100.0%)	45 (100.0%)	38 (100.0%)	49 (100.0%)
	Month 12	43 (84.3%)	30 (66.7%)	33 (86.8%)	43 (87.8%)
Serotype 6A	Month 1	50 (98.0%)	45 (100.0%)	38 (100.0%)	48 (98.0%)
	Month 12	44 (86.3%)	44 (97.8%)	35 (92.1%)	49 (100.0%)
Serotype 6B	Month 1	51 (100.0%)	45 (100.0%)	38 (100.0%)	49 (100.0%)
	Month 12	48 (94.1%)	41 (91.1%)	36 (94.7%)	46 (93.9%)
Serotype 6C	Month 1	50 (98.0%)	45 (100.0%)	35 (92.1%)	46 (93.9%)
	Month 12	36 (70.6%)	42 (93.3%)	25 (65.8%)	39 (79.6%)
Serotype 7F	Month 1	51 (100.0%)	45 (100.0%)	38 (100.0%)	49 (100.0%)
	Month 12	47 (92.2%)	45 (100.0%)	37 (97.4%)	49 (100.0%)
Serotype 9V	Month 1	51 (100.0%)	45 (100.0%)	38 (100.0%)	49 (100.0%)
	Month 12	44 (86.3%)	44 (97.8%)	38 (100.0%)	47 (95.9%)
Serotype 14	Month 1	51 (100.0%)	45 (100.0%)	38 (100.0%)	49 (100.0%)
	Month 12	48 (94.1%)	42 (93.3%)	29 (76.3%)	46 (93.9%)
Serotype 18C	Month 1	51 (100.0%)	45 (100.0%)	38 (100.0%)	49 (100.0%)
	Month 12	47 (92.2%)	43 (95.6%)	34 (89.5%)	47 (95.9%)
Serotype 19A	Month 1	51 (100.0%)	45 (100.0%)	38 (100.0%)	49 (100.0%)
	Month 12	43 (84.3%)	44 (97.8%)	33 (86.8%)	46 (93.9%)
Serotype 19F	Month 1	51 (100.0%)	45 (100.0%)	38 (100.0%)	49 (100.0%)
	Month 12	48 (94.1%)	36 (80.0%)	34 (89.5%)	47 (95.9%)
Serotype 23F	Month 1	51 (100.0%)	45 (100.0%)	38 (100.0%)	49 (100.0%)
	Month 12	47 (92.2%)	44 (97.8%)	36 (94.7%)	48 (98.0%)

OPA, opsonophagocytic assay; PCV13, 13-valent pneumococcal conjugate vaccine; PHiD-CV, 10-valent pneumococcal non-typeable *Haemophilus influenzae* protein D conjugate vaccine

Supplementary Table 3. Anti-pneumococcal antibody concentrations (95% CI) before and after the booster dose (per protocol set)

Serotype	3+1 schedule (Czech Republic)				2+1 schedule (Slovakia)			
	PHiD-CV group N = 53		PCV13 group N = 45		PHiD-CV group N = 39		PCV13 group N = 49	
	N	GMC (95% CI)	N	GMC (95% CI)	N	GMC (95% CI)	N	GMC (95% CI)
Serotype 1								
Month 0	51	0.22 (0.17; 0.27)	45	0.34 (0.28; 0.42)	38	0.27 (0.21; 0.34)	49	0.53 (0.45; 0.63)
Month 1	51	3.42 (2.72; 4.30)	45	2.54 (2.10; 3.09)	38	3.56 (2.75; 4.61)	49	4.31 (3.57; 5.20)
Month 12	48	0.40 (0.31; 0.51)	45	0.28 (0.23; 0.35)	38	0.41 (0.32; 0.54)	49	0.44 (0.37; 0.52)
Serotype 3								
Month 0	50	0.17 (0.12; 0.23)	45	0.24 (0.17; 0.33)	36	0.16 (0.12; 0.21)	49	0.35 (0.26; 0.47)
Month 1	48	1.12 (0.93; 1.35)	45	0.83 (0.64; 1.08)	34	1.10 (0.79; 1.54)	47	1.17 (0.90; 1.51)
Month 12	47	0.23 (0.17; 0.31)	44	0.20 (0.15; 0.28)	33	0.27 (0.18; 0.40)	47	0.25 (0.19; 0.33)
Serotype 4								
Month 0	51	0.31 (0.25; 0.39)	45	0.3 (0.24; 0.37)	38	0.37 (0.29; 0.49)	49	0.36 (0.30; 0.43)
Month 1	51	2.03 (1.72; 2.40)	45	2.77 (2.23; 3.45)	38	1.97 (1.53; 2.53)	49	2.58 (2.10; 3.17)
Month 12	48	0.21 (0.17; 0.25)	45	0.23 (0.18; 0.30)	38	0.17 (0.13; 0.21)	49	0.25 (0.20; 0.31)
Serotype 5								
Month 0	51	0.3 (0.25; 0.36)	45	0.31 (0.26; 0.38)	37	0.37 (0.30; 0.45)	49	0.34 (0.29; 0.40)
Month 1	51	1.24 (1.04; 1.48)	45	1.39 (1.15; 1.67)	38	1.28 (1.04; 1.59)	49	1.65 (1.34; 2.03)
Month 12	48	0.32 (0.25; 0.40)	45	0.29 (0.23; 0.37)	38	0.35 (0.29; 0.43)	49	0.33 (0.28; 0.40)
Serotype 6A								
Month 0	51	0.27 (0.20; 0.35)	45	0.54 (0.45; 0.65)	38	0.24 (0.17; 0.34)	49	0.68 (0.54; 0.86)
Month 1	51	6.29 (4.38; 9.04)	45	5.82 (4.83; 7.02)	37	4.60 (2.87; 7.37)	49	7.75 (6.27; 9.58)
Month 12	48	0.90 (0.64; 1.26)	45	0.63 (0.50; 0.80)	38	0.87 (0.64; 1.18)	49	1.15 (0.91; 1.45)
Serotype 6B								
Month 0	51	0.71 (0.58; 0.88)	45	0.37 (0.29; 0.47)	38	0.53 (0.40; 0.71)	49	0.24 (0.19; 0.31)
Month 1	51	13.07 (10.15; 16.83)	45	5.17 (4.39; 6.09)	38	10.25 (7.28; 14.42)	48	5.17 (4.02; 6.66)
Month 12	48	1.18 (0.87; 1.60)	45	0.71 (0.56; 0.90)	38	1.06 (0.80; 1.41)	49	0.85 (0.67; 1.10)
Serotype 7F								
Month 0	51	0.68 (0.54; 0.85)	45	0.77 (0.64; 0.93)	38	0.64 (0.51; 0.80)	49	0.72 (0.58; 0.88)
Month 1	51	3.34 (2.83; 3.93)	45	4.34 (3.71; 5.08)	38	2.78 (2.26; 3.42)	49	4.36 (3.55; 5.36)
Month 12	48	0.60 (0.50; 0.73)	45	0.69 (0.56; 0.85)	38	0.48 (0.40; 0.57)	49	0.71 (0.58; 0.87)
Serotype 9V								
Month 0	51	0.31 (0.24; 0.39)	45	0.26 (0.21; 0.31)	38	0.26 (0.21; 0.33)	49	0.29 (0.23; 0.38)
Month 1	51	2.08 (1.75; 2.48)	45	1.72 (1.34; 2.20)	38	2.11 (1.69; 2.63)	49	2.18 (1.76; 2.71)
Month 12	47	0.35 (0.27; 0.47)	45	0.25 (0.20; 0.33)	38	0.31 (0.25; 0.39)	49	0.37 (0.29; 0.48)
Serotype 14								
Month 0	51	1.84 (1.38; 2.45)	45	1.91 (1.46; 2.49)	38	1.83 (1.28; 2.61)	49	1.92 (1.39; 2.65)
Month 1	51	5.69 (4.64; 6.97)	45	7.60 (5.93; 9.75)	38	5.94 (4.75; 7.45)	49	9.16 (7.40; 11.34)
Month 12	48	0.87 (0.66; 1.13)	45	0.93 (0.67; 1.29)	38	0.94 (0.69; 1.28)	49	1.17 (0.91; 1.49)
Serotype 18C								
Month 0	50	0.46 (0.34; 0.62)	44	0.23 (0.18; 0.30)	38	0.43 (0.32; 0.58)	49	0.28 (0.22; 0.35)
Month 1	51	2.67 (2.13; 3.34)	45	2.00 (1.62; 2.47)	38	2.23 (1.73; 2.88)	49	2.22 (1.83; 2.70)
Month 12	48	0.28 (0.22; 0.35)	45	0.21 (0.16; 0.29)	38	0.30 (0.22; 0.39)	49	0.32 (0.26; 0.40)
Serotype 19A								
Month 0	51	0.33 (0.25; 0.42)	45	0.62 (0.45; 0.86)	38	0.38 (0.25; 0.57)	49	0.67 (0.55; 0.83)
Month 1	51	9.66 (6.70; 13.94)	45	9.72 (7.67; 12.32)	38	9.45 (6.80; 13.15)	49	8.22 (6.47; 10.45)
Month 12	48	1.30 (0.92; 1.82)	45	1.11 (0.78; 1.58)	38	1.11 (0.86; 1.44)	49	1.31 (1.00; 1.72)
Serotype 19F								
Month 0	51	1.83 (1.37; 2.44)	45	0.73 (0.59; 0.90)	38	2.03 (1.50; 2.74)	49	0.95 (0.80; 1.13)
Month 1	51	18.35 (13.70; 24.57)	45	7.65 (6.22; 9.40)	38	16.49 (11.94; 22.77)	49	9.36 (7.56; 11.60)
Month 12	48	1.90 (1.34; 2.70)	45	0.91 (0.68; 1.21)	38	1.72 (1.28; 2.30)	49	1.23 (0.98; 1.54)

Serotype	3+1 schedule (Czech Republic)				2+1 schedule (Slovakia)			
	PHiD-CV group		PCV13 group		PHiD-CV group		PCV13 group	
	N = 53		N = 45		N = 39		N = 49	
	N	GMC (95% CI)	N	GMC (95% CI)	N	GMC (95% CI)	N	GMC (95% CI)
Serotype 23F								
Month 0	51	0.41 (0.31; 0.55)	45	0.28 (0.22; 0.36)	38	0.34 (0.25; 0.46)	49	0.20 (0.15; 0.26)
Month 1	51	5.56 (4.18; 7.39)	45	4.35 (3.25; 5.82)	38	5.01 (3.56; 7.04)	49	3.62 (2.76; 4.76)
Month 12	48	0.47 (0.32; 0.70)	45	0.37 (0.27; 0.53)	38	0.37 (0.27; 0.53)	49	0.66 (0.41; 1.06)

GMC, geometric mean concentration; PCV13, 13-valent pneumococcal conjugate vaccine;

PHiD-CV, 10-valent pneumococcal non-typeable *Haemophilus influenzae* protein D

conjugate vaccine

Supplementary Table 4. Percentage of subjects with IgG geometric mean concentrations ≥ 0.35 $\mu\text{g/mL}$ 1 and 12 months after the PCV13 booster dose (per protocol set)

Serotype	Timing	3+1 schedule (Czech Republic)		2+1 schedule (Slovakia)	
		PHiD-CV N = 51	PCV13 N = 45	PHiD-CV N = 38	PCV13 N = 49
Serotype 1	Month 1	51 (100.0%)	45 (100.0%)	38 (100.0%)	49 (100.0%)
	Month 12	28 (54.9%)	17 (37.8%)	25 (65.8%)	30 (61.2%)
Serotype 3	Month 1	46 (90.2%)	39 (86.7%)	32 (84.2%)	45 (91.8%)
	Month 12	17 (33.3%)	9 (20.0%)	14 (36.8%)	14 (28.6%)
Serotype 4	Month 1	51 (100.0%)	45 (100.0%)	38 (100.0%)	49 (100.0%)
	Month 12	8 (15.7%)	11 (24.4%)	7 (18.4%)	15 (30.6%)
Serotype 5	Month 1	51 (100.0%)	44 (97.8%)	38 (100.0%)	49 (100.0%)
	Month 12	25 (49.0%)	17 (37.8%)	21 (55.3%)	26 (53.1%)
Serotype 6A	Month 1	49 (96.1%)	45 (100.0%)	35 (92.1%)	49 (100.0%)
	Month 12	39 (76.5%)	38 (84.4%)	32 (84.2%)	47 (95.9%)
Serotype 6B	Month 1	51 (100.0%)	45 (100.0%)	38 (100.0%)	47 (95.9%)
	Month 12	42 (82.4%)	36 (80.0%)	34 (89.5%)	43 (87.8%)
Serotype 7F	Month 1	51 (100.0%)	45 (100.0%)	38 (100.0%)	49 (100.0%)
	Month 12	41 (80.4%)	38 (84.4%)	30 (78.9%)	44 (89.8%)
Serotype 9V	Month 1	51 (100.0%)	43 (95.6%)	38 (100.0%)	48 (98.0%)
	Month 12	24 (47.1%)	14 (31.1%)	13 (34.2%)	25 (51.0%)
Serotype 14	Month 1	51 (100.0%)	45 (100.0%)	38 (100.0%)	49 (100.0%)
	Month 12	42 (82.4%)	37 (82.2%)	33 (86.8%)	46 (93.9%)
Serotype 18C	Month 1	51 (100.0%)	45 (100.0%)	38 (100.0%)	49 (100.0%)
	Month 12	17 (33.3%)	10 (22.2%)	14 (36.8%)	21 (42.9%)
Serotype 19A	Month 1	51 (100.0%)	45 (100.0%)	38 (100.0%)	49 (100.0%)
	Month 12	42 (82.4%)	39 (86.7%)	36 (94.7%)	48 (98.0%)
Serotype 19F	Month 1	51 (100.0%)	45 (100.0%)	38 (100.0%)	49 (100.0%)
	Month 12	46 (90.2%)	40 (88.9%)	37 (97.4%)	47 (95.9%)
Serotype 23F	Month 1	50 (98.0%)	44 (97.8%)	38 (100.0%)	49 (100.0%)
	Month 12	32 (62.7%)	21 (46.7%)	18 (47.4%)	30 (61.2%)

PCV13, 13-valent pneumococcal conjugate vaccine; PHiD-CV, 10-valent pneumococcal non-typeable *Haemophilus influenzae* protein D conjugate vaccine